

Volatility of the US stock market and business strategy during COVID-19

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Abstract

This paper aims to measure the impact of COVID-19 and changes in economic policy uncertainty (EPU) on the US stock market volatility using Event Study, Stepwise Regression, and Vector Autoregression (VAR) models. The impact has been examined under three contexts namely inside the US, outside the US and the whole world. This study finds negative cumulative abnormal returns of the market in response to the announcement of new cases and death records whereas the market reacts positively to the announcement of the financial bailout. The negative and significant impact of reported deaths and cases on the volatility of stock indices indicates that the US stock market is highly sensitive to the COVID-19. EPU has a significant and positive impact on the volatility of the US stock market. The COVID-19, EPU, and volatility variables are cointegrated and move in a unidirectional way. This study considers the presence of a long-run relationship among the variables and concludes that the US stock market is more exposed to COVID-19 compared with the rest of the world.

KEYWORDS

COVID-19, economic policy uncertainty, event study, granger causality, regression model, VAR, volatility

JEL CLASSIFICATION

G10, G14, G15, G18

1 | INTRODUCTION

Stock market indices react very rapidly to new events. Earlier studies have mentioned several significant events for which stock market returns have responded, including natural disasters (Lee & Chen, 2020), news announcements (Hussain and Ben Omrane, 2020), major sporting events (Curatola et al., 2016), environmental events (Guo et al., 2020), unexpected political shocks (Hillier & Loncan, 2019), and terrorist attacks (Goel et al., 2017). Stock market indices may also react to pandemic diseases, such as inter alia, the Ebola outbreak (Ichev & Marinč, 2018), and the Severe Acute Respiratory Syndrome outbreak (Chen et al., 2007). Yet despite this, there seems to be no research that has attempted to examine the effects of pandemic diseases on stock indices.

Starting from Wuhan city in China, the coronavirus disease (COVID-19) has so far infected 12,896,697 people, taken away 568,586 valuable lives, and paralyzed the financial system across the world (WHO, 2020). At the time of writing, United Kingdom, France, Spain, Iran, Italy, China, and United States are among the most heavily affected countries (Ayittey et al., 2020; Ayittey et al., 2021; Dhar et al., 2020). Gates (Gates, 2020) has termed the behavior of COVID-19 as “the once-in-a-century pathogen.” The pandemic has had a very significant effect on the real economic activity, and the extent of its impact on the macroeconomy is still unknown. COVID-19 has led to more damages to the economy of a country than other natural and human-caused disasters such as climate change, global nuclear conflicts, and localized conflicts (Goodell, 2020).

Considering these numerous effects, the literature on the financial impacts of COVID-19 has started to grow rapidly (Akhtaruzzaman

beginning, it turned negative after the first period and started to be positive after the second period. However, after the fourth period, the response is shown stable. It is evident that stock markets react negatively to the increasing number of confirmed cases and death cases. As the cases started to get stable, the volatility tends to follow the same pattern (Ashraf, 2020).

Since 1987, the time of the first modern market crash, the stock trading was briefly halted for 30 min only once in 1997. In order to deal with the risk contagions in the stock market, the circuit breaker has surprisingly been triggered four times within 10 days in March 2020. Although the price did not bounce back to its previous state during the financial crisis of 2001 and 2008, this time it was expected that if the disease could have been controlled, the reversion of asset price would have happened very soon. Although S&P 500 dropped by 30% from January 1, to March 24, 2020, within the very next week, the losses recovered by 20%. While the financial crisis of 2008 was mainly caused by the financial stress caused by the banking sector, the COVID-19 crisis started outside the financial sector. For these reasons, the stimulus package of the US government was not only directed to reverse the impacts of the crisis on the public sector but also on the private sector without any objection from the voters. Economists predict that the US economic growth will slow down by 2.8 percentage points as compared to the reduction of 1.7% in 2009 (BBC, 2020). The COVID-19 has not only paralyzed the US stock market but also the global economy. As regards the equity market volatility (EMV) tracker level, the value of COVID-19 in March 2020 was 60.46 compared with Ebola and MERS 2.06 in January 2015, Swine Flu 0.99 in May 2009, SARS 1.75 in August 2003, and Bird Flu 1.36 in November 1998 (Baker et al. 2020). The reason behind the heavier impact of COVID-19 pandemic on the stock market compared to earlier cases is probably related to the rapid growth of media outlets and the quick transformation of news industry. The USD 2 trillion stimulus package may help reverse this trend and underpin economic recovery. More stimulus, however, is required to reduce the high fiscal deficit (Kuttner, 2020).

5 | CONCLUSION

This paper attempted to measure the impact of COVID-19 and EPU on the US stock market using event study, stepwise regression, and VAR models. We observed that the market has changed significantly since COVID-19. The negative cumulative abnormal returns indicate a high sensitivity of the stock market toward announcements on new confirmed cases and deaths although the response of the market to the announcement of financial bailout was very optimistic. The results indicate that the COVID-19 confirmed cases and deaths both have a significantly negative impact on volatility. The uncertainty associated with the pandemic and the related economic losses has made the market extremely volatile and vulnerable. The study observes the existence of cointegration and long-run association among COVID-19, EPU, and volatility. Stock market is the index of a country's economy. Exactly when the stock market will return to stability and rationality is still difficult to predict. The announcement of a \$2 trillion stimulus

package and future policies to revive the economy may result in a fiscal shortfall. In order to comprehend the real impact of COVID-19 on the global economy, however, we need to wait for at least another quarter. This study only considered confirmed and death cases as COVID-19 variables using a relatively short time-frame. In future studies, researchers may consider other variables for a better understanding of this phenomenon.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

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ENDNOTE

* <https://www.worldometers.info/coronavirus/#countries>

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